How Plants Respond to Signals

Biology 2e chapter 9 reading connection | Teacher guide / classroom preparation

Learning target

Does a growth signal change stem elongation compared with its carrier alone?

Timing and materials

Setup plus daily observations for one to two weeks; grow seedlings in advance.

Dwarf pea seedlings of one documented variety, pots and soil, tray and grow light, ruler, labels, pipettes, educational gibberellic acid preparation and its matched carrier.

Classroom route

1. Grow same-age dwarf peas until shoots can be measured. Randomly assign at least three plants to each treatment and label each plant.

2. Use an educational GA3 preparation at the supplier-approved concentration and application volume for the chosen dwarf variety; follow its handling instructions. Apply an equal volume of matched carrier to control plants. Do not invent a stock formulation.

3. Measure stem height before treatment and daily for seven days. Keep water, light, soil and age comparable; rotate pot positions. Track each plant separately rather than treating repeated measurements as independent plants.

Pilot the physical procedure with your own equipment. Follow school handling and disposal rules and the purchased kit instructions. The virtual numerical settings are not an equipment prescription.

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Hybrid or substitute assignment

Assign all virtual, all physical, or specified samples in each mode. The classroom entry button is available during collection. Students complete the same prediction and evidence questions. Record classroom time series on the worksheet; the online classroom form stores endpoint measurements and notes. Keep sources and measurement scales separate.

Guided virtual sequence

Students prepare each sample, inspect the apparatus or classify specimens, then explicitly collect data. All assigned trials are required before exploration. Reload pauses the clock and restores saved work. Downloads retain source labels; no automatic grade is submitted.

Expected reasoning

Model treatment plants elongate more than carrier controls. Compare changes from baseline. The endpoint is a whole-plant response; it cannot identify a receptor or prove all steps of a signaling pathway.

Feedback and scoring

Criterion	Points	Evidence
Prediction and controls	0-2	Testable claim; comparable samples and controlled variables.
Observations	0-2	Complete, labeled records with units and source.
Interpretation	0-2	Correct calculations or categorical comparisons; claim tied to evidence.
Limitations	0-2	Specific uncertainty and a useful follow-up investigation.

Score reasoning and data quality, not agreement of physical results with the simulation. Manual tallies are student observations; a matching total does not prove every category is correct. Do not replace student counts with the model answer. The optional Sordaria extension retains classification feedback.

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Model limitations

A whole-plant response model, not a direct receptor assay. Response depends on genotype and growing conditions. Model growth is illustrative and does not establish a classroom dose.

Troubleshooting

Confirm the preparation steps are finished before starting or collecting. Corn and fly activities require manual tallies summing to the sample size plus review; optional specimen marks do not classify or tally automatically. Chromosome models require all four gametes plus a reviewed tally; cell measurements require a focused 40x view; gels require staining and three size estimates; fluorescence labs require the viewer. Use the visible Collect data and continue button to advance.

For physical work, preserve unexpected observations, check the control and instrument setup, then repeat with fresh material where appropriate. Never replace measured results with the virtual answer.

Evidence and optional classroom resources

University plant physiology investigation

https://schulte.faculty.unlv.edu/BIO442/Lab11DwarfPeas.pdf

External resources remain on their publishers' websites. This packet contains original text and tables, not supplier worksheets. Chapter numbers are reading connections. No supplier or textbook publisher endorsement is implied.

Teacher pilot checklist

Confirm availability and preparation time; test controls; check units and observation windows; verify handling/disposal; choose class grouping and accessible alternatives; keep notes for revision.